MH4110 Partial Differential Equations

Chapter 5 - Fourier Series

Outline

The goal of this chapter is to find the coefficients in a Fourier series.

- Fourier sine series
- Fourier cosine series
- Full Fourier series
- Convergence of Fourier series

Dirichlet Boundary Conditions – Heat equation

Theorem

The heat equation

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \ t > 0, \\ &u(0,t) = u(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{1}$$

has the series solution:

$$u(x,t) = \sum_{n=1}^{\infty} A_n \exp\Big(- \Big(\frac{n\pi}{l}\Big)^2 kt\Big) \sin \frac{n\pi x}{l}, \tag{2}$$

provided that ϕ has the sine Fourier expansion

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l}. \tag{3}$$

Dirichlet Boundary Conditions – Wave equation

Theorem

The wave equation

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l,\ t > 0, \\ &u(0,t) = u(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l, \end{aligned} \tag{4}$$

has the following series solution:

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l}, \tag{5}$$

as long as the initial data can be expanded into sine Fourier series

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l}, \quad \psi(x) = \sum_{n=1}^{\infty} \frac{n\pi c}{l} B_n \sin \frac{n\pi x}{l}. \tag{6}$$

Neumann Boundary Conditions – Heat equation

Theorem

The heat equation

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \ t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{7}$$

has the series solution:

$$u(x,t) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big) \cos\frac{n\pi x}{l}, \tag{8}$$

provided that ϕ has the cosine Fourier expansion

$$\phi(x) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l}. \tag{9}$$

Neumann Boundary Conditions – Wave equation

Theorem

The wave equation

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \ t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l, \end{aligned} \tag{10}$$

has the following series solution:

$$u(x,t) = \frac{A_0}{2} + \frac{B_0}{2}t + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \cos \frac{n\pi x}{l}, \tag{11}$$

as long as the initial data can be expanded into cosine Fourier series

$$\phi(x) = \frac{1}{2}A_0 + \sum_n A_n \cos \frac{n\pi x}{l}, \quad \psi(x) = \frac{1}{2}B_0 + \sum_n \frac{n\pi c}{l} B_n \cos \frac{n\pi x}{l}. \tag{12}$$

§5.1 The coefficients: Fourier sine series

Motivation: We have found Fourier series of several types, and the Fourier coefficients need to be determined by the initial condition(s).

Fourier sine series:

We revisit the series

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l}, \quad 0 < x < l. \tag{13}$$

It is a series of trigonometric sines, termed as a Fourier sine series. To find the coefficients A_n, let us first study the properties of the sines

$$\left\{ \sin \frac{\pi x}{l}, \sin \frac{2\pi x}{l}, \cdots, \sin \frac{n\pi x}{l}, \cdots \right\}. \tag{14}$$

- Recall that they are eigenfunctions of the eigenproblem:

$$-X''(x) = \lambda X(x), \quad X(0) = X(l) = 0.$$

§5.1 The coefficients: Fourier sine series

- They are periodic functions. A function ϕ that is defined for $-\infty < x < \infty$ is called periodic if there is a number $p > 0$ such that $\phi(x+p) = \phi(x)$ for all x. A number p for which this is true is called a **period** of $\phi(x)$. For instance, $\sin \omega x$ is $\frac{2\pi}{\omega}$-periodic. The period of $\sin \frac{n\pi x}{l}$ is $\frac{2l}{n}$.
- For each n, $\sin \frac{n\pi x}{l}$ is an odd function of x for all $x \in (-\infty, \infty)$.
- They are mutually orthogonal, namely,

$$\int_0^l \sin \frac{n\pi x}{l} \sin \frac{m\pi x}{l} dx = \begin{cases} l/2, & \text{if } n = m, \\ 0, & \text{if } n \neq m. \end{cases} \tag{15}$$

- They are linearly independent.

§5.1 The coefficients: Fourier sine series

- Recall the trigonometric identity:

$$\sin a \sin b = \frac{\cos(a-b) - \cos(a+b)}{2}.$$

We find that for $m, n \geq 1$,

$$\int_0^l \sin\frac{n\pi x}{l} \sin\frac{m\pi x}{l} dx = \frac{1}{2}\int_0^l \cos\frac{(m-n)\pi x}{l} dx - \frac{1}{2}\int_0^l \cos\frac{(m+n)\pi x}{l} dx.$$

It is clear that if $m = n$, the integral value is $l/2$, and it is zero when $m \neq n$.

§5.1 The coefficients: Fourier sine series

- We now determine the coefficients. Multiplying both sides of (13) by $\sin \frac{m\pi x}{l}$ and integrating the resulting equation over the interval $(0, l)$, we find from the orthogonality (15) that

$$\int_0^l \phi(x) \sin \frac{m\pi x}{l} dx = \sum_{n=1}^{\infty} A_n \int_0^l \sin \frac{n\pi x}{l} \sin \frac{m\pi x}{l} dx = A_m \frac{l}{2}.$$

This implies

$$A_m = \frac{2}{l} \int_0^l \phi(x) \sin \frac{m\pi x}{l} dx, \quad m = 1, 2, \cdots . \tag{16}$$

Therefore, **if** $\phi(x)$ **has an expansion** (13)**, then the coefficient must be given by** (16)**.**

§5.1 The coefficients: Fourier sine series

Theorem 1: Fourier sine series-Complete solution formula

Consider the heat equation:

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \quad t > 0, \\ &u(0,t) = u(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{17}$$

where k, l are positive constants and ϕ is a given function. It has the solution

$$u(x,t) = \sum_{n=1}^{\infty} A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big) \sin\frac{n\pi x}{l}, \tag{18}$$

where

$$A_n = \frac{2}{l}\int_0^l \phi(x)\sin\frac{n\pi x}{l}dx, \quad n = 1, 2, \cdots. \tag{19}$$

Remark: These are the only possible coefficients in (18). However, whether (18) is in fact valid with these values of the coefficients is a question of convergence.

§5.1 The coefficients: Fourier sine series

Example 1

Expand $\phi(x) = 1, x \in (0, l)$ in Fourier sine series.

Example 3

Expand $\phi(x) = x, x \in (0, l)$ in Fourier sine series and sketch the graph.

Solution: We need to evaluate

$$A_n = \frac{2}{l}\int_0^l x \sin\frac{n\pi x}{l}dx, \quad n = 1, 2, \cdots, \tag{22}$$

by integration by parts. More precisely,

$$\begin{aligned} A_n &= \frac{2}{l}\int_0^l x \sin\frac{n\pi x}{l}dx = -\frac{2}{n\pi}\int_0^l x\, d\cos\frac{n\pi x}{l} \\ &= -\frac{2}{n\pi}x\cos\frac{n\pi x}{l}\Big|_0^l + \frac{2}{n\pi}\int_0^l \cos\frac{n\pi x}{l}dx = \frac{2l}{n\pi}(-1)^{n+1}. \end{aligned}$$

Therefore, we have the Fourier sine expansion in $(0, l)$:

$$x = \frac{2l}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi x}{l} = \frac{2l}{\pi}\Big(\sin\frac{\pi x}{l} - \frac{1}{2}\sin\frac{2\pi x}{l} + \frac{1}{3}\sin\frac{3\pi x}{l}\cdots\Big). \tag{23}$$

§5.1 The coefficients: Fourier sine series

We plot the function $\phi(x) = x$ in $0 < x < l = 3$, and partial sum $S_m(x)$ of the first m terms of the Fourier sine series (29).

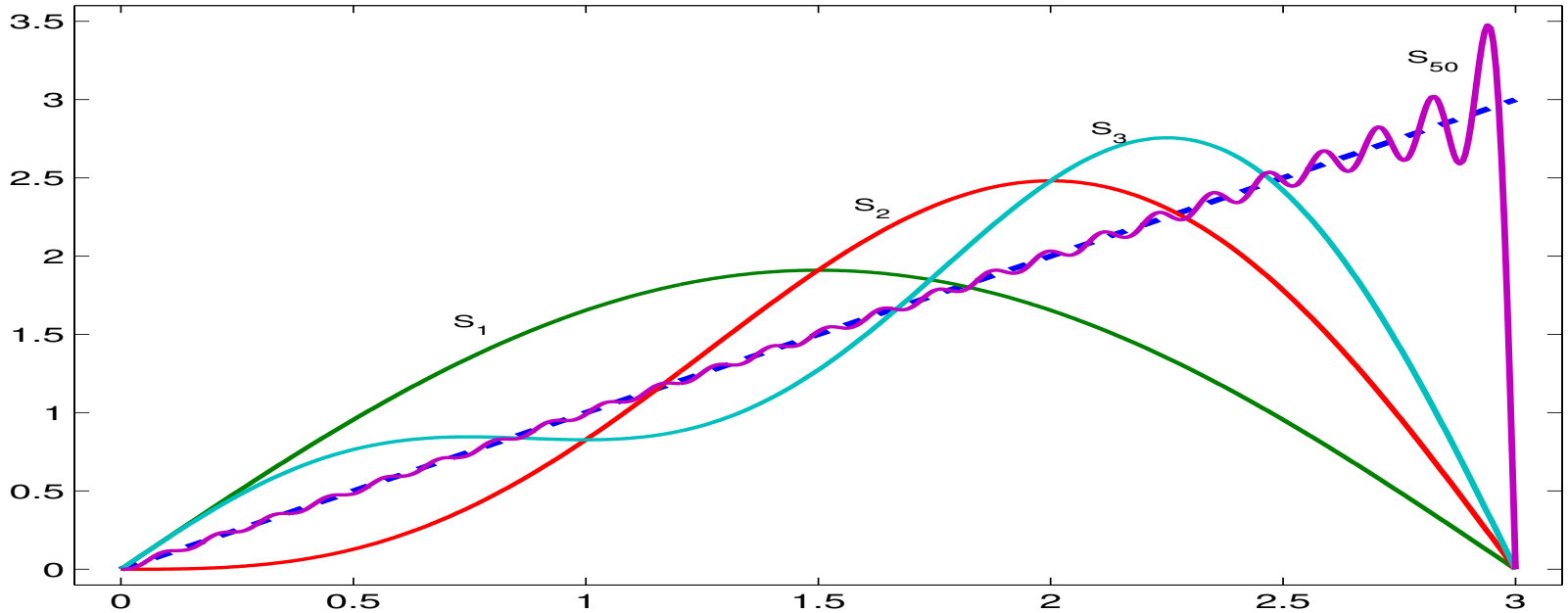

Figure: Fourier sine series of x (marked by dash line) and S_1, S_2, S_3, S_{50}.

We observe the oscillation occurs at the right endpoint (in general, at discontinuity), which is known as the **Gibb's phenomenon.**

§5.1 The coefficients: Fourier cosine series

In the study of wave and heat equations with Neumann boundary conditions in $(0, l)$, we have to consider the series of the form

$$\phi(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{l}, \tag{24}$$

which is referred to as a Fourier cosine series of $\phi(x)$ in $(0, l)$.

The system

$$\left\{1, \cos \frac{\pi x}{l}, \cos \frac{2\pi x}{l}, \cdots, \cos \frac{n\pi x}{l}, \cdots\right\}, \tag{25}$$

shares similar properties as the sines we discussed previously, but we notice that the cosines are even functions.

Moreover, we have the **orthogonality**:

$$\begin{aligned} &\int_0^l \cos \frac{n\pi x}{l} \cos \frac{m\pi x}{l} dx = 0, \quad \text{if} \ \ m \neq n; \\ &\int_0^l 1\, dx = l; \quad \int_0^l \left(\cos \frac{n\pi x}{l}\right)^2 dx = \frac{l}{2}, \quad n = 1, 2, \cdots. \end{aligned} \tag{26}$$

§5.1 The coefficients: Fourier cosine series

Thanks to the orthogonality, multiplying both sides of (24) by $\cos\frac{m\pi x}{l}$ and integrating the resulted identity over $(0, l)$, we find that

$$\begin{aligned}\int_0^l \phi(x)\cos\frac{m\pi x}{l}dx = \frac{A_0}{2}\int_0^l \cos\frac{m\pi x}{l}dx \\ + \sum_{n=1}^{\infty} A_n \int_0^l \cos\frac{n\pi x}{l}\cos\frac{m\pi x}{l}dx,\end{aligned} \tag{27}$$

which implies

$$A_m = \frac{2}{l}\int_0^l \phi(x)\cos\frac{m\pi x}{l}dx, \quad m = 0, 1, 2, \cdots. \tag{28}$$

At this point, we remark that to get a uniform expression (28) for all the coefficients we put the $\frac{1}{2}$ in front of the constant term in (24).

§5.1 The coefficients: Fourier cosine series

Theorem 2: Fourier cosine series-Complete solution formula

Consider the heat equation:

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \quad t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{29}$$

where k, l are positive constants and ϕ is a given function. It has the solution

$$u(x,t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big) \cos\frac{n\pi x}{l}, \tag{30}$$

where

$$A_n = \frac{2}{l}\int_0^l \phi(x)\cos\frac{n\pi x}{l}dx, \quad n = 0, 1, 2, \cdots. \tag{31}$$

Remark: These are the only possible coefficients in (30). However, whether (30) is in fact valid with these values of the coefficients is a question of convergence.

§5.1 The coefficients: Fourier cosine series

Example 2

Expand $\phi(x) = 1, x \in (0, l)$ in Fourier cosine series.

Example 4

Expand $\phi(x) = x, x \in (0, l)$ in Fourier cosine series and sketch the graph.

Solution: We need to evaluate

$$A_n = \frac{2}{l}\int_0^l x\cos\frac{n\pi x}{l}dx, \quad n = 0, 1, 2, \cdots. \tag{32}$$

For $n = 0$, we have

$$A_0 = \frac{2}{l}\int_0^l x dx = l.$$

For $n > 0$, we derive from integration by parts that

$$\begin{aligned} A_n &= \frac{2}{l}\int_0^l x\cos\frac{n\pi x}{l}dx = \frac{2}{n\pi}\int_0^l x\, d\sin\frac{n\pi x}{l} \\ &= \frac{2}{n\pi}x\sin\frac{n\pi x}{l}\Big|_0^l - \frac{2}{n\pi}\int_0^l \sin\frac{n\pi x}{l}dx \\ &= \frac{2l}{n^2\pi^2}\cos\frac{n\pi x}{l}\Big|_0^l = \frac{2l}{n^2\pi^2}\big((-1)^n - 1\big). \end{aligned}$$

§5.1 The coefficients: Fourier cosine series

Therefore, we have

$$A_{2k} = 0, \quad k = 1, 2, \cdots; \quad A_{2k+1} = -\frac{4l}{(2k+1)^2\pi^2}, \quad k = 0, 1, 2, \cdots.$$

Finally, we obtain the Fourier cosine expansion:

$$\begin{aligned} x &= \frac{l}{2} - \frac{4l}{\pi^2}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2}\cos\frac{(2k+1)\pi x}{l} \\ &= \frac{l}{2} - \frac{4l}{\pi^2}\left(\cos\frac{\pi x}{l} + \frac{1}{9}\cos\frac{3\pi x}{l} + \frac{1}{25}\cos\frac{5\pi x}{l}\cdots + \cdots\right), \quad x \in (0, l). \end{aligned} \tag{33}$$

This completes the solution.

§5.1 The coefficients: Fourier cosine series

We plot the function $\phi(x) = x$ in $0 < x < l = 3$, and partial sums $S_m(x)$.

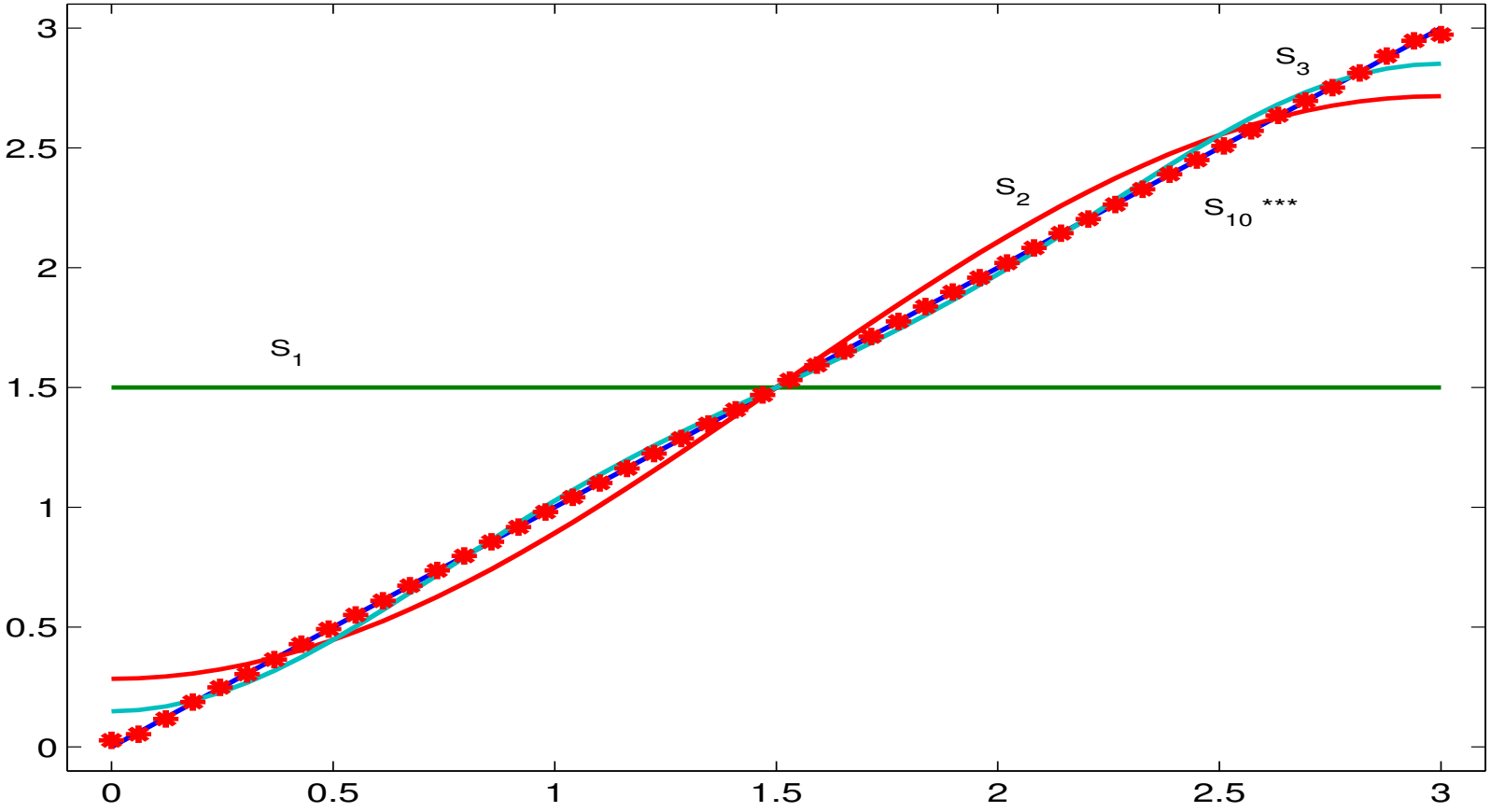

Figure: Fourier cosine series of x and S_1, S_2, S_3, S_{10} with S_{10} marked by "$*$".

§5.1 The coefficients: Full Fourier series

We have the orthogonality:

$$\int_{-l}^{l} \cos\frac{n\pi x}{l}\sin\frac{m\pi x}{l}dx = 0, \quad \text{for all } \ m, n;$$

$$\int_{-l}^{l} \cos\frac{n\pi x}{l}\cos\frac{m\pi x}{l}dx = 0, \quad \text{for all } \ m \neq n;$$

$$\int_{-l}^{l} \sin\frac{n\pi x}{l}\sin\frac{m\pi x}{l}dx = 0, \quad \text{for all } \ m \neq n.$$

In particular, we have

$$\int_{-l}^{l} 1\cdot\cos\frac{n\pi x}{l}dx = 0, \ \text{ for all } \ n \neq 0; \quad \int_{-l}^{l} 1\cdot\sin\frac{n\pi x}{l}dx = 0, \ \text{ for all } \ n.$$

Moreover, we have

$$\int_{-l}^{l} \cos^2\frac{n\pi x}{l}dx = l = \int_{-l}^{l} \sin^2\frac{n\pi x}{l}dx, \quad n \geq 1; \ \int_{-l}^{l} 1dx = 2l.$$

§5.1 The coefficients: Full Fourier series

Definition

The full Fourier series, or simply the Fourier series, of $\phi(x)$ on the interval $-l < x < l$, is defined as

$$\phi(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi x}{l} + B_n \sin \frac{n\pi x}{l} \right), \tag{34}$$

where

$$\begin{aligned} A_n &= \frac{1}{l} \int_{-l}^{l} \phi(x) \cos \frac{n\pi x}{l} dx, \quad n = 0, 1, 2, \cdots, \\ B_n &= \frac{1}{l} \int_{-l}^{l} \phi(x) \sin \frac{n\pi x}{l} dx, \quad n = 1, 2, \cdots. \end{aligned} \tag{35}$$

Example 5

Expand $\phi(x) = x, x \in [-l, l]$ in full Fourier series and sketch the graph.

Solution: We need to evaluate

$$A_n = \frac{1}{l}\int_{-l}^{l} x\cos\frac{n\pi x}{l}dx, \quad n = 0, 1, 2, \cdots$$

and

$$B_n = \frac{1}{l}\int_{-l}^{l} x\sin\frac{n\pi x}{l}dx, \quad n = 1, 2, \cdots .$$

For each A_n, since the integrand $x\cos\frac{n\pi x}{l}$ is an odd function in the interval $(-l, l)$, $A_n = 0$.

$$B_n = \frac{1}{l}\int_{-l}^{l} x\sin\frac{n\pi x}{l}dx = \left(\frac{-x}{n\pi}\cos\frac{n\pi x}{l} + \frac{1}{n^2\pi^2}\sin\frac{n\pi x}{l}\right)\Big|_{-l}^{l} = (-1)^{n+1}\frac{2l}{n\pi}.$$

§5.1 The coefficients: Full Fourier series

We plot the function $\phi(x) = x$ in $-3 = -l < x < l = 3$, and partial sums $S_m(x)$.

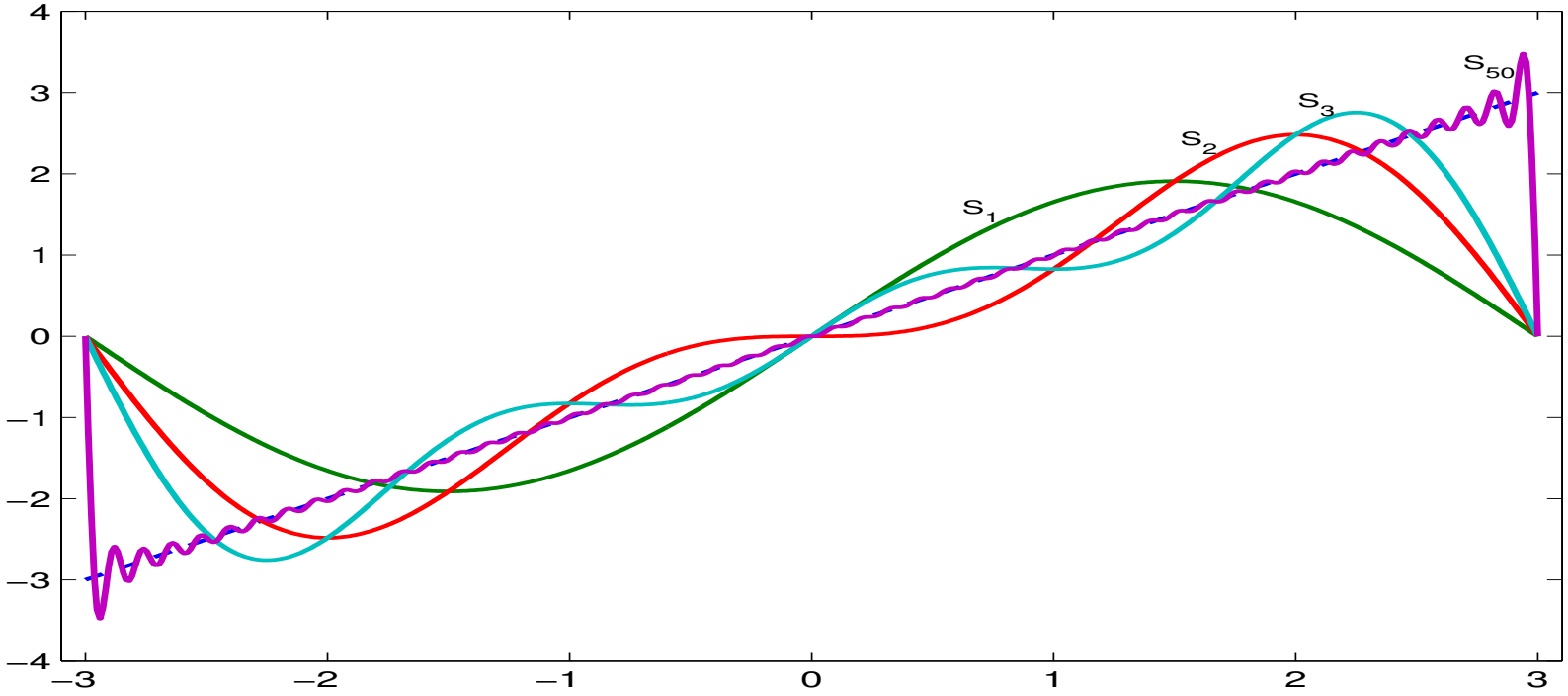

Figure: Fourier series of x (marked by dash-line) and S_1, S_2, S_3, S_{50}.

§5.1 The coefficients

Example 6

Solve the problem

$$
\begin{aligned}
&u_{tt} = c^2 u_{xx}, \quad 0 < x < l,\ t > 0, \\
&u(0,t) = u(l,t) = 0, \quad t > 0, \\
&u(x,0) = x, \quad u_t(x,0) = 0, \quad 0 < x < l,
\end{aligned}
\tag{36}
$$

Example 6 (Cont'd)

§5.1 The coefficients

Example A (not in the textbook)

Solve the problem

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l,\ t > 0, \\ &u(0,t) = u(l,t) = 0, \quad t > 0, \\ &u(x,0) = 0, \quad u_t(x,0) = 1, \quad 0 < x < l, \end{aligned} \tag{37}$$

Example A (Cont'd)

§5.4 Completeness

We state the basic theorems about the convergence of Fourier series in §5.4.

Three notions of convergence

(i) We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges to $f(x)$ **pointwisely**, if for each $x \in (a, b)$,

$$\lim_{N\to\infty} S_N(x) = f(x).$$

(ii) We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges to $f(x)$ **uniformly**, if

$$\lim_{N\to\infty} \max_{x\in[a,b]} |f(x) - S_N(x)| = 0.$$

(iii) We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges to $f(x)$ in the L^2-**sense** (or the **mean-square sense**), if

$$\lim_{N\to\infty} \int_a^b |f(x) - S_N(x)|^2 dx = 0.$$

§5.4 Completeness

Three notions of convergence (relations)

- Uniform convergence is **stronger** than the other two notions of convergence;
- Pointwise convergence and the L^2-convergence do not imply each other.

Example 1. Pointwise $\not\Rightarrow$ Uniform

Let $f_n(x) = (1-x)x^{n-1} = x^{n-1} - x^n$ on the interval $(0,1)$. Check the convergence of the series $\sum_{n=1}^{\infty} f_n(x)$ to $f(x) = 1$.

Solution: The partial sums are

$$\sum_{n=1}^{N} f_n(x) = \sum_{1}^{N} (x^{n-1} - x^n) = 1 - x^N \to 1 \quad \text{as} \quad N \to \infty.$$

This convergence is valid for each x. In words, the series converges pointwise to the function $f(x) = 1$. But the convergence is not uniform because $\max[1-(1-x^N)] = \max x^N = 1 \neq 0$ for every N. However, it does converge in mean-square since $\int_0^1 |x^N|^2 dx = \frac{1}{2N+1} \to 0$.

Theorem 2. Uniform Convergence

The Fourier (sine, cosine or full) series of $f(x)$, converges uniformly to $f(x)$ provided that

(i) for Fourier sine series: $f(x)$ and $f'(x)$ are continuous on $[0, l]$ and $f(0) = f(l) = 0$;

(ii) for Fourier cosine series: $f(x)$ and $f'(x)$ are continuous on $[0, l]$;

(iii) for full Fourier series: $f(x)$ and $f'(x)$ are continuous on $[-l, l]$ and $f(-l) = f(l)$.

Remark: The Fourier cosine series does not require the conditions at the endpoints, as the even extension will lead to $f(-l) = f(l)$.

Example: Examine the Fourier sine, cosine and Fourier full series of x that we discussed before.

Theorem 3. L^2-Convergence

The Fourier series converges to $f(x)$ in the mean-square sense in (a, b) provided only that $f(x)$ is any function for which

$$\int_a^b |f(x)|^2 dx$$

is finite, i.e., $f \in L^2(a, b)$.

Remark: The condition is weak, which means the Fourier series of a large class of functions converge in L^2-sense. Notice that all continuous functions are in L^2.

Theorem 4. Pointwise Convergence of Classical Fourier Series

(i) The classical Fourier series (full or sine or cosine) converges to $f(x)$ pointwisely on (a, b) provided that $f(x)$ is a continuous function on $[a, b]$, and $f'(x)$ is piecewise continuous on $[a, b]$.

(ii) More generally, if f itself is only piecewise continuous on $[a, b]$ and $f'(x)$ is also piecewise continuous on $[a, b]$, then the Fourier series converges at every point $x \in (-\infty, \infty)$. Moreover, we have the convergence:

$$\sum_n A_n X_n(x) = \frac{1}{2}\big[f(x+) + f(x-)\big], \quad -\infty < x < \infty, \tag{38}$$

where $f(a\pm) = \lim_{x\to a^\pm} f(x)$. Here, $f(x)$ is the extension (periodic, odd periodic or even periodic) of f in the original interval, e.g., $(0, l)$ or $(-l, l)$.

Theorem 4. Pointwise Convergence of Classical Fourier Series

(i) The classical Fourier series (full or sine or cosine) converges to $f(x)$ pointwisely on (a, b) provided that $f(x)$ is a continuous function on $[a, b]$, and $f'(x)$ is piecewise continuous on $[a, b]$.

(ii) More generally, if f itself is only piecewise continuous on $[a, b]$ and $f'(x)$ is also piecewise continuous on $[a, b]$, then the Fourier series converges at every point $x \in (-\infty, \infty)$. Moreover, we have the convergence:

$$\sum_n A_n X_n(x) = \frac{1}{2}\big[f(x+) + f(x-)\big], \quad -\infty < x < \infty, \tag{39}$$

Thus, at a jump discontinuity, the series converges to the average of the limits from the right and from the left. In the case of the Fourier sine (or cosine) series on $(0, l)$, the extended function $f_{\text{ext}}(x)$ is the odd (or even) function of period $2l$. For the full series on $(-l, l)$, it is the periodic extension. The extension is piecewise continuous with a piecewise continuous derivative on $(-\infty, \infty)$.